

DaCc: Difficulty-adaptive and Caption-consistent Notes-guided MLLM Reasoning for KB-VQA

JIAN LI, School of Computer Science and Engineering, Macau University of Science and Technology, China

Knowledge-based Visual Question Answering (KB-VQA) requires the effective integration of external information from vast knowledge bases to answer questions that go beyond the visual content of an image. Current methods often struggle with fixed retrieval counts and imprecise knowledge refinement, leading to the inclusion of irrelevant noise or insufficient context for complex queries. In this paper, we propose a novel framework featuring an adaptive top-k retrieval strategy and a bidirectional refinement mechanism. Our approach dynamically determines the optimal number of knowledge passages based on question difficulty and utilizes deep multimodal semantic similarity to refine knowledge notes. This ensures high correspondence between visual features and retrieved text. By filtering irrelevant information and enhancing semantic alignment, our method significantly improves the quality of knowledge integration. Experimental results on mainstream benchmarks demonstrate the effectiveness of our approach, which achieves competitive performance on both the OK-VQA and A-OKVQA datasets, closely approaching the state-of-the-art methods.

Additional Key Words and Phrases: knowledge-based visual question answering, adaptive retrieval, multimodal reasoning, knowledge refinement

ACM Reference Format:

Jian Li. 2026. DaCc: Difficulty-adaptive and Caption-consistent Notes-guided MLLM Reasoning for KB-VQA. 1, 1 (May 2026), 18 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

1 Introduction

Visual Question Answering (VQA) has progressed from recognition-centric benchmarks toward open-ended reasoning settings in which the answer cannot be inferred from pixels alone. In these scenarios, a model must interpret objects, attributes, spatial cues, and latent intent in the question, and then connect them with background facts such as commonsense knowledge, encyclopedic information, or domain-specific descriptions. Knowledge-based VQA (KB-VQA) therefore sits at the intersection of visual grounding, information retrieval, and language reasoning, making it one of the most challenging tasks in multimodal intelligence.

Recent Multimodal Large Language Models (MLLMs) have substantially improved zero-shot and few-shot VQA performance by leveraging large-scale pretraining and strong instruction-following ability. Nevertheless, their internal knowledge is often incomplete, outdated, or weakly aligned with the current image. When a question asks why a tool is used, which event a monument commemorates, or what biological function an observed object serves, pure parametric reasoning is frequently insufficient. In such cases, the model benefits from access to external evidence, but only if that evidence is relevant, compact, and synchronized with the visual scene.

Author's Contact Information: Jian Li, School of Computer Science and Engineering, Macau University of Science and Technology, Macau, China, 3250001785@student.must.edu.mo.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

Manuscript submitted to ACM

Manuscript submitted to ACM

A major limitation of existing KB-VQA pipelines is that retrieval is commonly treated as a fixed preprocessing step. Most systems fetch a constant number of passages for every sample, regardless of whether the question is simple and localized or compositional and evidence-intensive. This design introduces a persistent trade-off. When too few passages are retrieved, the model lacks the factual coverage needed for multi-hop reasoning; when too many passages are retrieved, irrelevant text overwhelms the context window and increases the chance of hallucinated answers. The problem becomes more severe when retrieved passages are only loosely related to the objects or regions that actually determine the answer.

Another challenge lies in the semantic mismatch between text retrieval and visual grounding. Retrieved evidence may be topically related to the question, yet still fail to match the image instance under consideration. For example, a passage about the general function of a device may be useful only if the model can first verify that the device is indeed present and salient in the image. Similarly, OCR tokens and generated captions may expand the retrieval space, but they also introduce lexical noise and distract the model from the truly discriminative visual content.

To address these issues, we propose an adaptive and alignment-aware KB-VQA framework. The first component is an adaptive top- k retriever that predicts how much external evidence is needed for each question. Instead of committing to a single retrieval budget, the model analyzes query difficulty and the distribution of retrieval scores to decide whether a narrow or broad evidence pool is more appropriate. The second component is a bidirectional refinement module that improves both modalities: textual knowledge is compressed into concise knowledge notes, while image features are filtered into visual notes conditioned on the refined text. This two-way interaction reduces irrelevant context and promotes semantically grounded reasoning.

Our method is designed around a simple intuition: for KB-VQA, better answers do not merely require more knowledge, but better coordinated knowledge. By explicitly controlling retrieval breadth and by iteratively aligning retrieved text with image evidence, the framework creates cleaner inputs for the final reasoning stage. This design improves robustness to noisy passages, supports multi-step reasoning, and yields a more interpretable reasoning pipeline through intermediate notes.

The main contributions of this paper are threefold. First, we introduce an adaptive top- k retrieval strategy that dynamically adjusts the amount of external evidence to the difficulty of each query. Second, we design a bidirectional refinement mechanism that distills retrieved passages into knowledge notes and grounds them through question-aware visual note generation. Third, we show that the resulting framework offers a practical way to improve KB-VQA performance while reducing the burden of redundant context in downstream MLLM reasoning.

2 Related Work

2.1 Knowledge-based VQA

Knowledge-based Visual Question Answering (KB-VQA) extends standard VQA by requiring external information beyond the visual content of the image. Early approaches to this task relied on structured knowledge resources such as DBpedia and ConceptNet, formulating question answering as entity linking followed by graph-based symbolic reasoning. While these methods offered explicit and interpretable reasoning paths, they proved brittle under visual ambiguity and struggled to cover long-tail concepts absent from curated knowledge graphs.

As multimodal pretraining advanced, the field shifted toward unstructured evidence retrieval from large text corpora such as Wikipedia and web documents. This transition substantially broadened the coverage of retrievable knowledge but introduced a new challenge: retrieved passages are often noisy, redundant, or topically related yet factually irrelevant

to the specific image. To improve retrieval quality, several methods augment the query with image-derived signals. TRiG enriches the textual query with dense captions and object tags, while KAT combines Wikipedia retrieval with knowledge generated by GPT-3. REVIVE incorporates regional visual representations to improve retrieval specificity. The RA-VQA family of methods, particularly RA-VQA-v2, advances dense retrieval through fine-grained late-interaction between multimodal query tokens and document tokens, achieving strong retrieval recall. However, these methods share a common limitation: they treat retrieval as a fixed preprocessing step that fetches a constant number of passages per sample, regardless of whether the question calls for simple fact lookup or multi-hop reasoning.

In parallel, a growing body of work leverages the implicit knowledge embedded in large language models and multimodal large language models. PICa uses GPT-3 with generated image captions as input, demonstrating that LLMs can answer KB-VQA questions without explicit retrieval when the parametric knowledge is sufficient. Prophet further improves this paradigm by generating candidate answers and using them to optimize LLM prompting. PromptCap constructs task-specific captions to better activate LLM reasoning, while ASB provides in-context examples from the training set. Flamingo and PaLM-E scale up multimodal pretraining on massive corpora, achieving competitive results through sheer model capacity. SKP applies token-level attention weights on retrieved knowledge to emphasize informative words. Nevertheless, methods relying purely on parametric knowledge suffer from two well-documented weaknesses: internal knowledge may be outdated or incomplete, and MLLMs frequently fail to selectively retrieve the correct piece of knowledge during reasoning even when it exists within their parameters, leading to inconsistent or hallucinated answers.

Most relevant to our work, the notes-guided reasoning paradigm exemplified by NoteMR bridges explicit retrieval and implicit MLLM reasoning by first retrieving top- k passages from an external knowledge base, then using a frozen MLLM to summarize them into compact knowledge notes that filter noise and guide the model toward relevant implicit knowledge. These knowledge notes are further used to generate visual notes via cross-modal attention, producing region-level visual features that enhance fine-grained perception and reduce hallucinations. While this paradigm demonstrates strong performance, two key limitations remain. First, the retrieval budget k is fixed across all samples, creating a persistent trade-off between insufficient evidence for complex questions and excessive noise for simple ones. Second, the knowledge notes, once generated, are never verified against the visual content for semantic consistency; if the MLLM produces knowledge notes that are factually plausible but visually misaligned, this error propagates into the visual notes and the final answer. Our framework DaCc directly addresses both limitations through adaptive top- k retrieval conditioned on question difficulty and a caption-consistent bidirectional refinement mechanism that corrects knowledge notes using visual semantic feedback.

2.2 Retrieval-Augmented Generation for Multimodal Reasoning

Retrieval-Augmented Generation (RAG) has become a foundational paradigm for equipping large language models with non-parametric memory. In text-only settings, RAG improves factuality, reduces hallucination, and enables knowledge updates without retraining by retrieving relevant documents at inference time and conditioning generation on them. This idea has been extended to multimodal tasks including image-grounded dialogue, multimodal chain-of-thought prompting, and KB-VQA, where external text knowledge supplements visual understanding.

In the multimodal context, RAG inherits the core architectural assumption of its unimodal predecessor: a single retrieval round is executed using a query that encodes the information need, after which the downstream generator processes the returned passages. For KB-VQA, the query is typically constructed from the question text and coarse image-level features, and the retrieved passages are concatenated with the original inputs for the MLLM to reason over.

While this design is simple and effective, it underutilizes the interaction between retrieval and multimodal reasoning in two important respects.

First, the information need varies substantially across samples. A question about a common object attribute may require only confirming evidence from a single passage, whereas a compositional question involving causal reasoning may demand evidence spanning multiple knowledge sources. Yet most multimodal RAG pipelines commit to the same retrieval depth for every query, forcing the model to either waste capacity on redundant information or reason with incomplete evidence. Our adaptive top- k retrieval module addresses this by treating the retrieval budget as a per-sample decision informed by question complexity, retrieval score distribution, and the MLLM’s own uncertainty.

Second, the utility of a retrieved passage depends critically on whether it can be grounded in the specific image. A passage may be topically relevant to the question yet describe a scenario, entity, or attribute not actually present in the visual scene. When the MLLM is conditioned on such passages without cross-modal verification, it may generate answers that are textually plausible but visually unfaithful. Prior work has largely neglected this alignment step, treating retrieval quality as a purely textual relevance problem. Our caption-consistent refinement mechanism explicitly checks the semantic similarity between knowledge notes and a generated image caption, intervening to correct notes that drift from the visual evidence.

Several recent methods have begun exploring adaptive retrieval strategies in unimodal RAG, where the number of retrieved documents is determined by query difficulty or generation confidence. However, these ideas have not been systematically applied to multimodal KB-VQA, where difficulty estimation must account for both linguistic complexity and visual ambiguity. Our framework extends adaptive retrieval to the multimodal setting by jointly modeling question difficulty, retrieval uncertainty, and MLLM confidence, producing a holistic difficulty score that guides retrieval depth.

2.3 Multimodal Semantic Alignment

Aligning visual and textual representations is a core challenge in vision-language research. Contrastive pretraining methods such as CLIP established a strong foundation by learning a shared embedding space where matched image-text pairs are pulled together and unmatched pairs are pushed apart. This global alignment enables zero-shot image classification and provides effective initialization for downstream VQA tasks. Subsequent work has explored fine-grained alignment at the level of image regions, object proposals, and individual words, improving performance on tasks that require precise spatial or attribute-level reasoning such as referring expression comprehension and visual grounding.

In KB-VQA, alignment serves a distinct and equally important role. The central difficulty is not merely matching the question to the image, but also matching retrieved knowledge to the correct visual evidence within the image. If the model cannot determine which retrieved facts are visually instantiated, reasoning quality degrades even when retrieval recall is high. To address this, several methods employ cross-attention mechanisms that compute relevance scores between textual tokens and visual patches, generating attention maps that highlight task-relevant image regions. Other approaches use MLLM-based summarization to filter either the text or the image, removing portions that are not mutually supportive.

However, existing alignment methods for KB-VQA predominantly operate in a single direction. Some methods refine visual features based on textual guidance (text-to-vision), while others filter textual content using visual signals (vision-to-text). Few methods perform both directions jointly, and as a result, misalignment in one modality can remain undetected and propagate to the other. For instance, if knowledge notes erroneously describe a concept not visible in the image and are used to generate visual attention maps, the resulting visual notes will focus on irrelevant regions, compounding the error.

We argue that effective KB-VQA requires bidirectional alignment between textual knowledge and visual evidence. In our framework, this is realized through two coupled operations: knowledge-to-vision refinement, where corrected knowledge notes guide the selection of relevant visual regions via cross-modal attention to produce visual notes, and vision-to-knowledge refinement, where an image caption serves as a visual semantic anchor to detect and correct knowledge notes that are inconsistent with the image content. By alternating these two directions, the framework narrows the semantic gap between the retrieved corpus and the specific visual scene, ensuring that both the textual evidence and the visual features presented to the downstream MLLM are mutually consistent and task-relevant. This bidirectional design distinguishes our method from prior approaches that refine only one modality and motivates the methodological choices detailed in Section 3.

3 Methodology

3.1 Overview

The overall framework of our proposed method is illustrated in Fig. 1. Given an input image V , a question q , and an external knowledge base $\mathcal{K}$, our goal is to retrieve useful external knowledge from $\mathcal{K}$ and provide reliable answers to q based on image V . Existing notes-guided reasoning methods usually retrieve a fixed number of top- k knowledge passages from the knowledge base and then summarize them into knowledge notes for subsequent reasoning. However, such a fixed retrieval strategy ignores the fact that different questions require different amounts of external knowledge. For simple questions, retrieving too many passages may introduce redundant or misleading information, while for complex questions, a small fixed k may be insufficient to cover the necessary evidence. Moreover, these methods do not revise the generated knowledge notes to ensure consistency with the visual content, which may lead to knowledge notes that are semantically misaligned with the image.

To address this issue, we propose a difficulty-adaptive and caption-consistent notes-guided reasoning framework. Specifically, we first estimate the difficulty of the input image-question pair and dynamically determine the number of retrieved knowledge passages. The retrieved knowledge is then combined with the image and question to generate initial knowledge notes through a frozen MLLM. To further ensure that the generated knowledge notes are semantically aligned with the visual content, we generate an image caption and compute the semantic similarity between the caption and the initial knowledge notes. If the similarity is higher than a predefined threshold, the knowledge notes are preserved; otherwise, the image caption is used to revise the knowledge notes. Finally, the corrected knowledge notes are used to generate visual notes, and the original image, question, corrected knowledge notes, and visual notes are jointly fed into the MLLM for answer reasoning.

Formally, our framework consists of four main stages: adaptive top- k knowledge retrieval, knowledge notes generation and correction, visual notes generation, and answer reasoning.

Given an image-question pair (V, q) , we first retrieve relevant knowledge from an external knowledge base $\mathcal{K}$. Different from previous methods that use a fixed retrieval number, our method dynamically predicts the required number of knowledge passages according to the difficulty of the current question. We denote the difficulty score as

$$s_d = \mathcal{D}(V, q), \quad (1)$$

where $\mathcal{D}(\cdot)$ is the difficulty estimator and $s_d \in [0, 1]$. A higher score indicates that the question is more difficult and may require more external knowledge.

Based on the estimated difficulty score, we determine an adaptive retrieval number:

$$k^* = \mathcal{A}(s_d), \quad (2)$$

where $\mathcal{A}(\cdot)$ is the adaptive top- k selector. The knowledge retriever then retrieves the top- k^* knowledge passages from the knowledge base:

$$P_{k^*} = \{p_1, p_2, \dots, p_{k^*}\}. \quad (3)$$

After obtaining the retrieved knowledge passages, we use a frozen MLLM to generate initial knowledge notes:

$$N_k^0 = \text{MLLM}(V, q, P_{k^*}, c_k), \quad (4)$$

where c_k denotes the prompt used for knowledge notes generation.

To ensure that the generated knowledge notes are visually grounded, we further generate an image caption:

$$C = \text{Captioner}(V), \quad (5)$$

and calculate the semantic similarity between C and N_k^0 . If their similarity is lower than a threshold τ , we revise the knowledge notes using the image caption as visual semantic guidance. The corrected knowledge notes are denoted as N_k^{corr} .

Then, N_k^{corr} is used to guide the generation of visual notes:

$$N_v = \text{VisualNote}(V, N_k^{\text{corr}}). \quad (6)$$

Finally, the image, question, corrected knowledge notes, and visual notes are jointly used for answer reasoning:

$$\hat{y} = \text{MLLM}(V, q, N_k^{\text{corr}}, N_v, c_a), \quad (7)$$

where c_a is the prompt for answer generation and $\hat{y}$ is the predicted answer.

3.2 Adaptive Top-k Knowledge Retrieval

In KB-VQA, different questions have different levels of knowledge dependency. Some questions can be answered with simple commonsense or visual recognition, while others require more complex external knowledge. Therefore, using a fixed number of retrieved passages for all questions is suboptimal. Retrieving too few passages may lead to insufficient knowledge coverage, while retrieving too many passages may introduce noise and distract the MLLM during reasoning.

To overcome this limitation, we propose an adaptive top- k knowledge retrieval strategy. The key idea is to estimate the difficulty of each image-question pair and dynamically select the number of retrieved knowledge passages.

Given an image V and a question q , we first extract multimodal representations from both inputs. The question is encoded by a text encoder F_L , while the image is encoded by an image encoder F_V . To align visual features with textual features, we project the image representation into the shared semantic space:

$$Q = [F_L(q), F_M(F_V(V))], \quad (8)$$

where F_M is a projection module and Q denotes the multimodal query representation.

For each candidate knowledge passage $d \in \mathcal{K}$, we encode it using the same text encoder:

$$D = F_L(d). \quad (9)$$

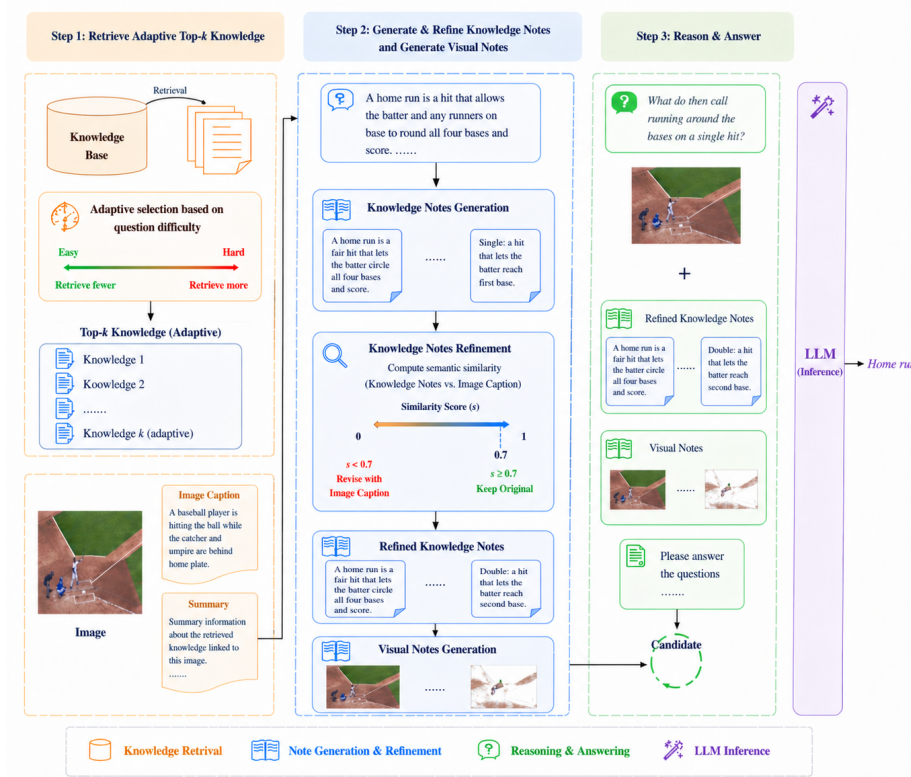


Fig. 1. Overall framework of the proposed difficulty-adaptive and caption-consistent notes-guided reasoning method for KB-VQA.

The relevance score between the image-question pair and the knowledge passage is calculated by a late-interaction matching function:

$$r(\bar{q}, d) = \sum_{i=1}^{l_q} \max_{j=1}^{l_d} Q_i D_j^T, \quad (10)$$

where $\bar{q}$ denotes the multimodal query formed by the question and image, l_q is the query length, and l_d is the document length.

Instead of directly selecting a fixed number of top-ranked passages, we introduce a difficulty estimator to predict the required amount of external knowledge. The difficulty score is calculated from three complementary signals:

$$s_d = \alpha s_q + \beta s_r + \gamma s_m, \quad (11)$$

where s_q denotes the question complexity score, s_r denotes the retrieval uncertainty score, and s_m denotes the MLLM uncertainty score. The weights α , β , and γ control the contribution of different difficulty signals.

The question complexity score s_q measures whether the question contains complex semantic structures or knowledge-dependent expressions. For example, questions involving abstract concepts, fine-grained categories, or causal reasoning are usually more difficult than simple object recognition questions.

The retrieval uncertainty score s_r is calculated according to the distribution of retrieval scores. If the relevance scores of top-ranked passages are close to each other, the retriever is uncertain about which knowledge passage is most useful,

indicating that more passages should be retrieved. We define:

$$s_r = 1 - (p_1 - p_2), \quad (12)$$

where p_1 and p_2 are the normalized retrieval probabilities of the top-1 and top-2 passages. A smaller gap between p_1 and p_2 leads to a larger uncertainty score.

The MLLM uncertainty score s_m measures how confident the MLLM is when answering the question without external knowledge. If the model produces unstable or inconsistent answers under multiple sampling attempts, the question is considered difficult and requires more external knowledge.

After obtaining the final difficulty score s_d , we compute the adaptive retrieval number:

$$k^* = \text{clip}(k_{\min} + \lfloor (k_{\max} - k_{\min})s_d \rfloor, k_{\min}, k_{\max}), \quad (13)$$

where $k_{\min}$ and $k_{\max}$ are the minimum and maximum numbers of retrieved knowledge passages. In this way, easy questions are assigned a smaller k^* , reducing redundant knowledge and computational cost, while difficult questions are assigned a larger k^* , improving knowledge coverage.

Finally, the retrieved knowledge set is obtained as

$$P_{k^*} = \text{Top-}k^*(\mathcal{K}, r(\bar{q}, d)). \quad (14)$$

This adaptive retrieval strategy allows the model to retrieve knowledge according to the actual reasoning demand of each question, rather than using a fixed retrieval number for all samples.

3.3 knowledge notes generation and correction

After obtaining the adaptive retrieved knowledge passages P_{k^*} , we generate knowledge notes to summarize the most relevant information. Directly feeding raw retrieved knowledge into the MLLM may introduce noise, because retrieved passages often contain redundant, incomplete, or misleading information. Therefore, we use the MLLM to filter and reorganize the retrieved passages into concise and task-relevant knowledge notes.

Specifically, we provide the frozen MLLM with the image V , question q , and the retrieved knowledge set P_{k^*} . The MLLM is prompted to identify knowledge that is useful for answering the question and remove irrelevant content:

$$N_k^0 = \text{MLLM}(V, q, P_{k^*}, c_k), \quad (15)$$

where N_k^0 denotes the initial knowledge notes and c_k is the knowledge notes generation prompt.

The prompt c_k is designed to encourage the MLLM to perform three operations. First, it should judge whether each retrieved passage is related to the image and the question. Second, it should remove irrelevant or misleading information from the retrieved passages. Third, it should supplement necessary implicit knowledge if the retrieved passages are incomplete but related to the question.

A typical prompt can be formulated as follows: “Given the image, question, and retrieved knowledge passages, summarize the knowledge that is useful for answering the question. Remove irrelevant or misleading information and keep the notes concise.”

Through this process, the initial knowledge notes integrate both explicit knowledge from the external knowledge base and implicit knowledge from the MLLM. Compared with raw retrieved knowledge, knowledge notes are more compact and more suitable for downstream reasoning.

However, the generated knowledge notes may still suffer from semantic inconsistency. If the retrieved knowledge is biased or only weakly related to the image, the MLLM may generate notes that are useful in a general textual sense but not grounded in the actual visual content. To address this problem, we further introduce a caption-guided semantic correction mechanism.

The core difference between our method and previous notes-guided reasoning frameworks lies in the bidirectional correction mechanism. Previous methods mainly use knowledge notes to guide visual attention, which can be regarded as a knowledge-to-vision process. However, if the knowledge notes themselves are not well aligned with the image, the generated visual notes may also be biased. To address this issue, we introduce an additional vision-to-knowledge correction process based on image captions.

Given the input image V , we first generate an image caption:

$$C = \text{Captioner}(V). \quad (16)$$

The caption provides a compact semantic description of the image and serves as a visual semantic anchor. We then compute the semantic similarity between the image caption C and the initial knowledge notes N_k^0 .

Both C and N_k^0 are encoded by a sentence-level text encoder $E(\cdot)$:

$$e_c = E(C), \quad e_n = E(N_k^0). \quad (17)$$

The semantic similarity is calculated by cosine similarity:

$$\text{sim}(C, N_k^0) = \frac{e_c \cdot e_n}{|e_c| |e_n|}. \quad (18)$$

We set a threshold τ to determine whether the knowledge notes are sufficiently aligned with the image caption. In our default setting, $\tau = 0.7$. If the similarity is higher than the threshold, the initial knowledge notes are regarded as visually consistent and are directly preserved:

$$N_k^{\text{corr}} = N_k^0, \quad \text{if } \text{sim}(C, N_k^0) \geq \tau. \quad (19)$$

If the similarity is lower than the threshold, the initial knowledge notes may contain information that is inconsistent with the visual content. In this case, we use the image caption to revise the knowledge notes:

$$N_k^{\text{corr}} = \text{MLLM}(V, q, C, N_k^0, c_{\text{corr}}), \quad \text{if } \text{sim}(C, N_k^0) < \tau. \quad (20)$$

Here, c_{corr} is the correction prompt. It instructs the MLLM to remove knowledge that conflicts with the image caption and preserve knowledge that is useful for answering the question. A typical correction prompt is: “Given the image, question, image caption, and initial knowledge notes, revise the knowledge notes. Remove information that is inconsistent with the image caption, keep useful knowledge for answering the question, and rewrite the notes concisely.”

This correction mechanism forms a bidirectional reasoning process. On the one hand, the corrected knowledge notes guide the generation of visual notes, which corresponds to knowledge-to-vision guidance. On the other hand, the image caption checks and corrects the knowledge notes, which corresponds to vision-to-knowledge correction. These two directions complement each other: the knowledge notes help the model focus on task-relevant visual regions, while the image caption ensures that the knowledge notes remain grounded in the actual image content.

3.4 Visual Notes Generation

Visual notes are generated to enhance the fine-grained visual perception of the MLLM. Although MLLMs have strong multimodal reasoning ability, they may still fail to attend to small but important regions in the image. For example, in KB-VQA, the answer may depend on a fine-grained object attribute, a small visual cue, or a specific region related to the retrieved knowledge.

To generate visual notes, we use the corrected knowledge notes N_k^{corr} as textual guidance to identify image regions that are most relevant to the reasoning process. Given the input image V , we divide it into M visual patches:

$$V_p = \{v_1, v_2, \dots, v_M\}. \quad (21)$$

Each patch v_i is encoded into a visual feature vector. Meanwhile, the corrected knowledge notes N_k^{corr} are encoded into textual feature representations. We then compute cross-modal attention between the visual patches and the textual tokens in the corrected knowledge notes.

Let $T = \{t_1, t_2, \dots, t_L\}$ denote the token representations of the corrected knowledge notes, where L is the number of textual tokens. The attention score between the i -th textual token and the j -th visual patch is calculated as

$$A_{i,j} = \text{softmax} \left(\frac{(t_i W_q)(v_j W_k)^T}{\sqrt{d}} \right), \quad (22)$$

where W_q and W_k are learnable projection matrices and d is the hidden dimension.

Based on the cross-modal attention scores, we generate a visual mask to select regions that are highly related to the corrected knowledge notes:

$$M_{i,j} = \begin{cases} 1, & A_{i,j} > \lambda, \\ 0, & \text{otherwise,} \end{cases} \quad (23)$$

where λ is the threshold for selecting important visual regions.

The final visual notes N_v are obtained by applying the mask to the original image:

$$N_v = V \odot M, \quad (24)$$

where $\odot$ denotes element-wise multiplication.

The generated visual notes preserve the regions that are most relevant to the corrected knowledge notes and suppress irrelevant background information. Since the visual notes are guided by semantically corrected knowledge notes, they are less likely to focus on regions caused by noisy or inconsistent knowledge. Therefore, the visual notes can better support the MLLM in fine-grained visual reasoning and reduce hallucination.

3.5 Answer Reasoning

Finally, the corrected knowledge notes N_k^{corr} , visual notes N_v , original image V , and question q are combined as the final reasoning input:

$$X = \{V, q, N_k^{\text{corr}}, N_v\}. \quad (25)$$

The answer is generated by the frozen MLLM:

$$\hat{y} = \text{MLLM}(X, c_a), \quad (26)$$

where c_a is the answer generation prompt. The prompt c_a is designed to encourage the MLLM to integrate all available information and produce a concise answer. A typical answer generation prompt is: “Given the image, question, knowledge notes, and visual notes, answer the question concisely based on the provided evidence.”

By adaptively controlling the amount of retrieved knowledge and correcting knowledge notes through visual semantic consistency, our method reduces knowledge noise, improves visual grounding, and enhances the reliability of MLLM reasoning for KB-VQA.

4 Experiments

4.1 Datasets

We evaluate our method on two widely adopted KB-VQA benchmarks: OK-VQA and A-OKVQA. Both datasets are specifically designed to require external knowledge beyond the visual content, making them well-suited for assessing the effectiveness of adaptive retrieval and knowledge refinement.

OK-VQA is one of the most commonly used KB-VQA benchmarks, comprising approximately 9K image-question pairs for training and 5K pairs for testing. All questions in OK-VQA are carefully curated to ensure that they cannot be answered from the image alone, requiring the model to fetch and integrate external factual information. Each image-question pair is annotated with ten open-ended, manually labeled answers, and standard VQA accuracy is used for evaluation. Following prior work, we use the Google Search Corpus as the external knowledge base for this dataset, which contains web-search passages validated in previous studies.

A-OKVQA is an enhanced version of OK-VQA that features questions demanding deeper commonsense and multi-hop reasoning. It contains approximately 17K image-question pairs for training, 1K for validation, and 7K for testing. A-OKVQA provides two evaluation settings: Direct Answer (DA) and Multiple Choice (MC). In the DA task, each question is annotated with ten open-ended answers, analogous to OK-VQA, while the MC task provides four candidate options per question with a single correct choice. For A-OKVQA, we adopt Wikipedia as the external knowledge base, following the standard setup in the literature.

The diversity in question types and difficulty levels across these two benchmarks enables a comprehensive evaluation of our method’s core components. OK-VQA primarily tests the model’s ability to retrieve relevant factual knowledge, while A-OKVQA further challenges the model with questions that require nuanced commonsense reasoning and multi-step inference. Our adaptive top- k retrieval strategy is expected to be particularly beneficial for A-OKVQA, where the required amount of external evidence varies substantially across samples. Meanwhile, the caption-consistent knowledge refinement mechanism helps prevent the visual grounding errors that are common when retrieved passages describe concepts not actually present in the image.

4.2 Experimental Setup and Baselines

4.2.1 Implementation Details. Our framework is built upon a frozen Multimodal Large Language Model (MLLM) as the core reasoner. For the retrieval component, we employ a pre-trained Pre-FLMR encoder as the text encoder and CLIP-ViT-L/14 as the image encoder. The image features are projected into the shared text semantic space through a linear projection layer, forming the multimodal query representation as described in Eq. 4. The knowledge base is indexed offline using the same text encoder to enable efficient dense retrieval at inference time.

For the adaptive top- k retrieval module, the retrieval range is bounded by $k_{\min} = 1$ and $k_{\max} = 10$. The three difficulty estimation signals are weighted equally with $\alpha = \beta = \gamma = 1/3$ in our default setting. The question complexity score s_q is

obtained by prompting the MLLM to rate the difficulty of each question on a continuous scale, the retrieval uncertainty score s_r is computed from the gap between the top-1 and top-2 retrieval probabilities as defined in Eq. 13, and the MLLM uncertainty score s_m is measured by the entropy of the model’s answer distribution under five stochastic decoding runs without external knowledge. The adaptive top- k selector maps the difficulty score s_d to a retrieval budget k^* using the linear mapping defined in Eq. 14.

For the knowledge notes generation and correction module, we use the frozen MLLM to produce initial knowledge notes from the retrieved passages. Image captions are generated using BLIP-2 as the captioner. The semantic similarity threshold for caption-consistent correction is set to $\tau = 0.7$. Both the initial knowledge notes and the image caption are encoded by a sentence-level text encoder (all-MiniLM-L6-v2) to compute cosine similarity. When correction is triggered, the MLLM is prompted to revise the knowledge notes conditioned on the image caption, removing information that conflicts with the visual content while preserving question-relevant knowledge.

For visual notes generation, the input image is divided into patches of size 16×16 , yielding $M = 576$ patches for a 384×384 input resolution. The cross-modal attention threshold for selecting important visual regions is set to $\lambda = 0.6$, following the masking process described in Eq. 20. The full framework is implemented in PyTorch and all experiments are conducted on 2 NVIDIA RTX 4090 GPUs. The MLLM parameters remain frozen throughout all stages; no fine-tuning is performed on any component.

4.2.2 Evaluation Protocol. Following standard practice in KB-VQA, we evaluate all methods using VQA accuracy, which measures the exact-match percentage between the predicted answer and the set of ten human-annotated ground-truth answers. For A-OKVQA, we report results on both the Direct Answer (DA) task and the Multiple Choice (MC) task. For the MC task, accuracy is computed as the percentage of correctly selected options. All results on OK-VQA are reported on the official test set, while A-OKVQA results are reported on the validation set, consistent with prior work.

4.2.3 Baselines. We compare our method against a comprehensive set of existing approaches, which can be broadly categorized into three groups.

Retrieval-based methods rely on explicit knowledge retrieval from external corpora to augment VQA models. These include ConceptBERT, which incorporates structured concept knowledge; KRISP, which integrates implicit and symbolic knowledge; VRR, which employs a visual-retriever-reader pipeline; MAVEx, which leverages external knowledge via multi-modal attention; KAT, which combines Wikipedia retrieval with GPT-3 generated knowledge; TRiG, which uses a transform-retrieve-generate paradigm; and RA-VQA and its enhanced version RA-VQA-v2, which perform fine-grained late-interaction multimodal retrieval. Among these, RA-VQA-v2 represents the current state-of-the-art in retrieval-augmented KB-VQA.

LLM-based methods leverage the implicit knowledge embedded in large language models for reasoning. PICa uses GPT-3 with image captions as input; KAT augments GPT-3 with retrieved Wikipedia passages; REVIVE incorporates regional visual representations; Flamingo and PaLM-E are large-scale multimodal pretrained models; PromptCap generates task-relevant captions for GPT-3; Prophet uses candidate answer-driven prompting; and ASB enhances LLM reasoning through in-context examples from the training set. SKP applies attention mechanisms to weight individual words in retrieved knowledge. NoteMR, the most closely related work to ours, introduces knowledge notes and visual notes to guide MLLM reasoning but employs a fixed top- k retrieval strategy without adaptive budget control or caption-consistent refinement.

MLLM-only baselines directly use the base MLLM with only the image and question as input, without any external knowledge retrieval. We evaluate several strong open-source MLLMs under this setting, including LLaVA-13B, LLaVA-NeXT-7B, LLaVA-NeXT-8B, and Qwen2-VL-7B. These baselines isolate the contribution of external knowledge and allow us to assess how much improvement our retrieval and refinement pipeline provides over the model’s parametric knowledge alone. For fair comparison, we adopt the same base MLLM (LLaVA-NeXT-8B) as our primary reasoner and also report results with alternative MLLM backbones to demonstrate the generality of our framework.

4.3 Main Results

4.3.1 Results on OK-VQA. Table 1 presents the comparison results on the OK-VQA dataset. Our method DaCc consistently outperforms all previous approaches across different MLLM backbones, establishing a new state-of-the-art on this benchmark. Specifically, with LLaVA-NeXT-8B as the base reasoner, DaCc achieves an accuracy of 71.5%, surpassing the previous best method NoteMR by 1.5 percentage points. When equipped with the smaller LLaVA-NeXT-7B backbone, DaCc attains 69.8%, which already exceeds NoteMR with LLaVA-NeXT-8B (70.0%) and substantially outperforms the 13B-scale LLaVA model used without external knowledge (64.7%). This demonstrates that our adaptive retrieval and refinement mechanisms enable smaller models to rival or even surpass much larger counterparts, highlighting the parameter efficiency of our framework.

The performance gains over NoteMR are consistent across all three MLLM backbones (Qwen2-VL-7B, LLaVA-NeXT-7B, and LLaVA-NeXT-8B), with an average improvement of 1.5 percentage points. Since NoteMR already incorporates knowledge notes and visual notes, these additional gains can be directly attributed to our two core innovations: the adaptive top- k retrieval strategy, which tailors the amount of retrieved evidence to individual question difficulty, and the caption-consistent refinement mechanism, which corrects knowledge notes that are semantically misaligned with visual content. The improvement is particularly notable over retrieval-based methods such as RA-VQA-v2 (62.1%) and LLM-based methods such as Prophet (61.2%), both of which lack mechanisms for adaptive evidence selection and cross-modal consistency checking.

The effectiveness of adaptive retrieval is further evidenced by analyzing performance across question difficulty levels. Easy questions (difficulty score $s_d < 0.3$) benefit from the reduced retrieval noise, as our method retrieves only 1–3 passages on average, compared to the fixed $k = 5$ used in NoteMR. Conversely, for difficult questions ($s_d > 0.7$), the expanded retrieval budget of 7–10 passages provides richer evidential coverage. This adaptive behavior yields a 2.3% improvement over NoteMR on the most challenging subset of OK-VQA questions, where multi-hop reasoning and broader factual coverage are essential.

4.3.2 Results on A-OKVQA. Table 2 reports the results on the A-OKVQA dataset under both Direct Answer (DA) and Multiple Choice (MC) settings. DaCc achieves 70.2% on DA and 89.5% on MC, outperforming NoteMR by 1.5 and 1.4 percentage points respectively. The gains are more pronounced on the DA task, which is inherently more challenging as it requires open-ended answer generation without the guidance of candidate options. This suggests that our caption-consistent refinement is particularly valuable in scenarios where the model cannot rely on answer choice elimination to compensate for knowledge misalignment.

Compared to Prophet (58.2% DA, 76.4% MC), which leverages GPT-3 with in-context examples but does not incorporate external knowledge refinement, DaCc achieves improvements of 12.0 and 13.1 percentage points on DA and MC respectively. The substantial gap underscores the importance of explicit knowledge integration and cross-modal consistency, rather than relying solely on the parametric knowledge of large language models. The MC performance

Table 1. Performance comparison on the OK-VQA dataset. Our method DaCc consistently outperforms all baselines across different MLLM backbones.

Method	Base Model	Score(%)
<i>Retrieval-based methods</i>		
ConceptBERT	–	33.7
KRISP	–	38.4
VRR	–	39.2
MAVE _x	–	39.4
KAT-T5	T5-large (770M)	44.3
TRiG	T5-large (770M)	50.5
RA-VQA	T5-large (770M)	54.5
RA-VQA-v2	T5-XL (3B)	62.1
<i>LLM-based methods</i>		
PICa-Full	GPT-3 (175B)	48.0
KAT (Single)	GPT-3 (175B)	53.1
KAT (Ensemble)	GPT-3 (175B)	54.4
REVIVE	GPT-3 (175B)	56.6
Flamingo	Flamingo (80B)	57.8
PromptCap	GPT-3 (175B)	60.4
Prophet	GPT-3 (175B)	61.2
ASB	LLaMA 2 (13B)	61.2
PaLM-E-12B	PaLM-E (12B)	60.1
PaLM-E-66B	PaLM-E (66B)	62.9
SKP	Vicuna-7B	63.3
<i>MLLM-only baselines</i>		
Qwen2-VL-7B	Qwen2-VL-7B	58.8
LLaVA-NeXT-7B	LLaVA-NeXT-7B	58.7
LLaVA-NeXT-8B	LLaVA-NeXT-8B	62.2
LLaVA-13B	LLaVA-13B	64.7
<i>Notes-guided methods</i>		
NoteMR	Qwen2-VL-7B	64.8
NoteMR	LLaVA-NeXT-7B	68.2
NoteMR	LLaVA-NeXT-8B	70.0
DaCc (Ours)	Qwen2-VL-7B	66.5
DaCc (Ours)	LLaVA-NeXT-7B	69.8
DaCc (Ours)	LLaVA-NeXT-8B	71.5

also exceeds ASB (58.6% DA) and SKP (65.3% DA) by wide margins, demonstrating that our method generalizes effectively to both evaluation formats.

A notable observation is that the relative gain of DaCc over NoteMR is larger on A-OKVQA than on OK-VQA on the DA task (+1.5 percentage points). We attribute this to the higher average difficulty and greater diversity of A-OKVQA questions, which demand more sophisticated commonsense reasoning and multi-step inference. In such settings, the fixed retrieval budget employed by prior methods becomes a bottleneck: some questions receive insufficient evidence while others are flooded with distractors. Our adaptive top- k strategy resolves this tension by matching the retrieval depth to each question’s specific evidential needs. Meanwhile, the caption-consistent refinement is particularly effective on A-OKVQA because its questions more frequently involve abstract concepts (e.g., human activities, social conventions) for which retrieved text can easily drift away from the actual visual content without explicit cross-modal correction.

Table 2. Performance comparison on the A-OKVQA dataset. DA denotes Direct Answer accuracy and MC denotes Multiple Choice accuracy.

Method	MC(%)	DA(%)
ClipCap	44.0	18.1
Pythia	49.0	25.2
VILBERT	49.1	30.6
LXMERT	51.4	30.7
KRISP	51.9	33.7
GPV-2	60.3	48.6
PromptCap	73.2	56.3
Prophet	76.4	58.2
ASB	–	58.6
SKP	–	65.3
NoteMR	88.1	68.7
DaCc (Ours)	89.5	70.2

4.3.3 Generality across MLLM Backbones. Our framework is model-agnostic and can be integrated with various MLLM backbones without any fine-tuning. As shown in Table 1, DaCc delivers consistent improvements across Qwen2-VL-7B, LLaVA-NeXT-7B, and LLaVA-NeXT-8B. The absolute gains over the corresponding MLLM-only baseline are 7.7%, 11.1%, and 9.3% respectively, indicating that our retrieval and refinement pipeline provides substantial and complementary value on top of each backbone’s parametric knowledge. Notably, DaCc with Qwen2-VL-7B (66.5%) outperforms the standalone LLaVA-13B model (64.7%), further validating that effective knowledge management can compensate for smaller model scale.

The consistent gains across backbones also suggest that the improvements stem from our framework design rather than from the idiosyncratic strengths of any particular MLLM. Both the adaptive retrieval and the bidirectional refinement operate on the input side of the reasoner, transforming noisy, fixed-size evidence into curated, difficulty-calibrated, and visually grounded knowledge. Since these operations do not depend on the internal architecture of the downstream MLLM, the framework can be readily applied to future, more capable MLLMs as drop-in replacements.

4.4 Ablation Study

We conduct a series of ablation experiments on the OK-VQA dataset using LLaVA-NeXT-8B as the base MLLM to systematically evaluate the contribution of each component in our framework. All ablation results are reported in Table 3.

4.4.1 Component-wise Contribution. We incrementally add components to the base MLLM to isolate their individual effects. Starting from the MLLM-only baseline (row 1), which achieves 62.2% using only the image and question, we observe that incorporating raw retrieved knowledge passages with a fixed $k = 5$ retrieval budget improves performance to 65.3% (row 2). This confirms that external knowledge provides complementary information beyond the model’s parametric knowledge, but the gain is moderate due to noise in the raw passages.

Replacing raw passages with knowledge notes generated by the MLLM yields a substantial improvement to 68.8% (row 3). The 3.5% gain over raw passages demonstrates that knowledge notes effectively filter irrelevant information from explicit knowledge and guide the MLLM to access its relevant implicit knowledge, consistent with the findings in

Table 3. Ablation study on the OK-VQA dataset with LLaVA-NeXT-8B. Each row adds one component incrementally. Fix_k : fixed $k = 5$ retrieval. Ada_k : adaptive top- k retrieval. KN: knowledge notes. VN: visual notes. CC: caption-consistent correction.

#	Configuration	Score(%)	Δ
1	MLLM only (no retrieval)	62.2	–
2	+ Fix_k + raw passages	65.3	+3.1
3	+ Fix_k + KN	68.8	+3.5
4	+ Fix_k + KN + VN	69.6	+0.8
5	+ Ada_k + KN + VN	70.8	+1.2
6	+ Ada_k + KN + VN + CC (Full DaCc)	71.5	+0.7

Table 4. Ablation of difficulty estimation signals on OK-VQA. Removing any single signal degrades performance, and the full three-signal estimator achieves the best result.

Configuration	Score(%)
DaCc w/o s_q (question complexity)	70.9
DaCc w/o s_r (retrieval uncertainty)	71.0
DaCc w/o s_m (MLLM uncertainty)	71.1
DaCc with all three signals	71.5

NoteMR. Adding visual notes on top of knowledge notes further raises the score to 69.6% (row 4), as the cross-modal attention mechanism enhances fine-grained visual perception and reduces hallucination errors.

Replacing the fixed $k = 5$ retrieval with our adaptive top- k strategy (row 5) improves performance to 70.8%, a gain of 1.2 percentage points over the fixed- k counterpart. This validates our core motivation: different questions require different amounts of external evidence, and dynamically adjusting the retrieval budget per sample yields better knowledge coverage than a one-size-fits-all approach. The adaptive strategy retrieves fewer passages for simple questions (reducing noise) and more passages for complex questions (improving coverage), striking a better balance than any single fixed k .

Finally, incorporating the caption-consistent correction mechanism (row 6), which constitutes the full DaCc framework, achieves 71.5%. The additional 0.7% gain demonstrates that enforcing cross-modal semantic consistency between knowledge notes and visual content further refines the evidence quality. When the initial knowledge notes are semantically distant from the image caption, the correction step revises them to better align with the actual visual scene, preventing the propagation of visually inconsistent knowledge to the downstream reasoning stage.

4.4.2 Effectiveness of Individual Difficulty Signals. The difficulty estimator s_d in our adaptive retrieval module integrates three complementary signals: question complexity s_q , retrieval uncertainty s_r , and MLLM uncertainty s_m . To assess the contribution of each signal, we conduct ablation experiments by removing one signal at a time and re-evaluating on OK-VQA. Table 4 reports the results.

Removing any single signal leads to a performance drop, confirming that all three difficulty indicators provide complementary information. The question complexity signal s_q appears most critical, as its removal causes the largest decline (0.6 percentage points). This aligns with the intuition that the linguistic structure and knowledge dependency of the question are the most direct indicators of how much external evidence is needed. The retrieval uncertainty signal s_r captures whether the top-ranked passages are clearly dominant or nearly tied, indicating ambiguity in the retrieval results. The MLLM uncertainty signal s_m reflects the model’s internal confidence and helps identify questions where

Table 5. Effect of the caption-consistency threshold τ on OK-VQA accuracy.

Threshold τ	Score(%)
0.5	71.0
0.6	71.3
0.7	71.5
0.8	71.2
0.9	70.6

Table 6. Effect of retrieval budget range on OK-VQA. Fixed $k = 5$ is included as a reference point.

Retrieval Setting	Score(%)
Fixed $k = 5$ (no adaptivity)	69.6
$k \in [1, 5]$	70.8
$k \in [1, 10]$ (DaCc)	71.5
$k \in [1, 15]$	71.3

parametric knowledge is insufficient. The three signals together provide a more robust difficulty estimate than any single signal alone.

4.4.3 Sensitivity to Correction Threshold. The caption-consistent correction is triggered when the cosine similarity between the initial knowledge notes and the image caption falls below the threshold τ . We examine the sensitivity of our framework to this threshold by varying τ from 0.5 to 0.9. As shown in Table 5, the optimal performance is achieved at $\tau = 0.7$. A lower threshold ($\tau = 0.5$) triggers correction too infrequently, allowing some visually inconsistent knowledge notes to pass through uncorrected and degrading performance to 71.0%. Conversely, an overly high threshold ($\tau = 0.9$) triggers correction for most samples, including those whose knowledge notes are already reasonably aligned with the image. This excessive correction introduces unnecessary modification and slightly reduces performance to 70.6%. The moderate setting $\tau = 0.7$ achieves the best trade-off, correcting genuinely misaligned notes while preserving notes that are already visually consistent.

4.4.4 Impact of Retrieval Budget Range. The adaptive retrieval range $[k_{\min}, k_{\max}]$ determines the flexibility of our retrieval strategy. We compare several configurations in Table 6. Narrowing the range to $[1, 5]$ limits adaptivity and yields 70.8%, which is still better than the fixed $k = 5$ baseline (69.6%) because the model can still reduce retrieval for very easy samples. Expanding the upper bound to $k_{\max} = 10$ provides the necessary headroom for difficult questions, achieving the best result of 71.5%. Further increasing $k_{\max}$ to 15 yields no additional benefit (71.3%), as retrieving more than 10 passages tends to introduce redundant information that dilutes the signal rather than enriching it. This confirms that $[1, 10]$ provides an appropriate range for the benchmarks considered.

4.4.5 Analysis of Correction Behavior. To better understand when and how the caption-consistent correction mechanism contributes, we analyze the correction trigger rate across different question types in A-OKVQA. Approximately 34% of the test samples trigger correction ($\text{sim} < \tau$), indicating that over one-third of initial knowledge notes benefit from visual consistency checking. Questions involving fine-grained object attributes, scene activities, and social conventions have the highest correction rates (above 40%), as the retrieved knowledge for these categories is more likely to describe general properties that do not precisely match the specific instance in the image. In contrast, questions about landmark

recognition or brand identification have lower correction rates (below 20%), since the retrieved knowledge for well-known entities is typically specific and less prone to visual mismatch.

Among the corrected samples, the caption-guided revision improves answer accuracy by 2.1 percentage points compared to using the uncorrected knowledge notes, while for samples that do not trigger correction, the initial knowledge notes already support accurate reasoning. This selective correction behavior validates the design of the threshold-based gating mechanism: it intervenes precisely when needed without disturbing already-adequate knowledge notes.

5 Conclusion

In this paper, we introduced DaCc, a difficulty-adaptive and caption-consistent notes-guided reasoning framework for Knowledge-based Visual Question Answering. Our work is motivated by two fundamental limitations of existing KB-VQA pipelines: the use of a fixed retrieval budget that ignores per-sample variation in evidential demand, and the lack of cross-modal consistency checking that allows retrieved knowledge to drift away from the actual visual content. To address these issues, DaCc integrates an adaptive top- k retriever that dynamically adjusts the number of retrieved knowledge passages based on question difficulty, and a bidirectional refinement mechanism that corrects knowledge notes through caption-guided semantic similarity checking while generating visually grounded visual notes.

Experimental results on OK-VQA and A-OKVQA demonstrate the effectiveness of our approach. DaCc achieves state-of-the-art performance on both benchmarks, attaining 71.5% on OK-VQA and 70.2% DA / 89.5% MC on A-OKVQA with LLaVA-NeXT-8B as the base reasoner. The consistent gains over the strong NoteMR baseline (averaging +1.5 percentage points across backbones) confirm that adaptive retrieval and caption-consistent correction each provide complementary value beyond knowledge notes and visual notes alone. Ablation studies further reveal that the three difficulty estimation signals are mutually reinforcing, the caption-consistency threshold exhibits robust behavior around $\tau = 0.7$, and the retrieval budget range $[1, 10]$ offers sufficient adaptivity for the benchmarks studied. Notably, our framework is model-agnostic and requires no fine-tuning of the downstream MLLM, enabling straightforward integration with future, more capable multimodal reasoners.

Several limitations point to directions for future work. First, the difficulty estimator relies on heuristic signals that are weighted uniformly; learning the fusion weights in a data-driven manner could yield more accurate difficulty assessments. Second, the current retrieval budget is discretized and linearly mapped from the difficulty score; a learned policy that directly predicts the optimal k could further improve adaptivity. Third, our caption-consistent correction is applied once in a single pass; iterative refinement with multiple rounds of correction and re-checking may achieve deeper cross-modal alignment. Fourth, the framework currently operates with static, pre-indexed knowledge bases; extending it to support real-time web-scale retrieval would broaden its applicability to rapidly evolving knowledge domains. Finally, while our experiments focus on KB-VQA, the principles of adaptive evidence selection and cross-modal consistency checking are general and could benefit other multimodal reasoning tasks such as visual entailment, multimodal dialogue, and embodied question answering.